Mitigating and Simulating Algorithmic Price Collusion

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Abstract

Algorithmic pricing agents trained via Q-learning reliably converge to tacit collusion, yet regulators currently lack effective tools to counteract this behavior when it appears in the real world. This paper asks whether a platform-imposed price threshold can meaningfully reduce collusion among competing Q-learning sellers. Drawing on the Calvano et al. framework and extending it with a dynamic platform learning mechanism, we simulate markets of two to five sellers under logit and winner-take-all demand. We find that a threshold set to exclude high-pricing sellers reduces collusion by up to 62% in five-seller logit markets, with consumer surplus rising by over 23%. The mechanism is most effective when the number of sellers meaningfully exceeds the platform's target eligible count. These results suggest that price thresholds represent an effective, market-compatible policy instrument—one that regulators could plausibly mandate without requiring direct access to a firm's pricing algorithms.

1 Introduction

In many of the world's largest online marketplaces, prices are no longer set by human sellers. Instead, automated pricing algorithms update prices by the minute or second, continuously optimizing for profit based on competitor prices, demand signals, and platform rules. On Amazon alone, many of its 600 million product listings are likely managed by algorithmic repricers [1] [2, p. 1339]. For consumers, this might seem like a recipe for fierce competition. In practice, the evidence suggests otherwise.

A growing empirical and theoretical literature documents that pricing algorithms can sustain prices far above competitive levels through tacit collusion, with no explicit coordination among sellers. Existing legal remedies are poorly suited to this form of harm, since antitrust doctrine requires an explicit agreement that tacit algorithmic coordination never produces. Prior platform-side interventions have reduced collusion but do so by compressing seller margins to the point of threatening market viability. By setting price caps or other controls, they reduce tacit collusion in a way that makes seller participation unprofitable. We explore more effective rules for platforms that reduce algorithmic collusion while preserving seller participation. Chapter 2 reviews the empirical evidence, theoretical foundations, and prior interventions in detail.

This paper makes two contributions toward filling this gap. The first is a theoretical framework formalizing a platform threshold mechanism as a solution to algorithmic collusion. The platform sets a maximum price threshold τ_t, excluding any seller priced above it from eligibility for a platform's "Buy Box." We derive the conditions under which this threshold is incentive-compatible for the platform, model the platform's dynamic learning problem when seller cost distributions are unobservable, and prove that this threshold will converge to a stable target. Critically, the mechanism preserves seller participation by design: sellers priced below the threshold continue to earn their normal demand share and can remain profitable. The second contribution is simulation evidence using Q-learning agents. This is similar to the experiments conducted by Calvano et al. [3], but extended to heterogeneous seller costs and multiple seller counts. We show that the threshold reduces the collusion index by 40–62% under logit demand, with corresponding consumer surplus gains of 23–25%, and that effectiveness is determined by the exclusion

margin $N - S_{\text{target}}$ rather than seller count alone. We further test robustness under winner-take-all demand, finding comparable collusion reductions but more variable consumer surplus effects, and identify the market conditions under which the threshold can reduce consumer welfare. The platform's threshold learning rule converges to a stable equilibrium in all primary configurations.

2 Literature Review

This chapter reviews four interconnected bodies of literature that motivate and contextualize the threshold mechanism developed in this paper: empirical evidence on algorithmic pricing in practice, the Q-learning framework for modeling tacit collusion, platform design interventions, and the legal gap that makes voluntary or regulatory platform action necessary.

2.1 Algorithmic Pricing in Practice

The prevalence and competitive consequences of algorithmic pricing have been documented across a range of market settings. Chen et al. [2] conducted the first large-scale empirical study of algorithmic pricing on Amazon, identifying over 500 sellers using automated repricing strategies among top-selling products. They found that algorithmic sellers are systematically more successful than human pricers: they win the Buy Box more frequently, respond faster to rival price changes, and are better at gaming platform visibility rules beyond price alone [2, p. 1340].

Musolff [4] extends this analysis to understand the mechanisms by which algorithmic repricers generate profits, showing that repricing algorithms create collusive cycles that drive prices higher for consumers. When sellers use algorithmic repricers, prices tend to follow periods of undercutting that are followed by jumps upward toward monopoly levels [4, pp. 2-3]. The mechanism is not coordination; it is rational best-response learning in a repeated game. Thanks to lower menu costs on Amazon that are around approximately 0.02%, this near-continuous repricing is essentially costless [5, p. 7]. Musolff estimates the median markup on Amazon at 9–18%, and argues that the presence of predictable cycling patterns is itself evidence of supra-competitive pricing. Competitive markets at marginal cost would exhibit no such cycles [5, p. 8].

Perhaps the most direct evidence that collusion is a feature rather than a bug of repricing software comes from the marketing materials of the repricing industry itself. Seller Snap, one of the leading Amazon repricers, openly advertises a “cooperative strategy” designed to “[keep] the price high, instead of racing to the bottom,” explaining that its algorithm

evaluates how competitors react to higher prices to identify the maximum sustainable price level [6], [7]. Repricer.com similarly markets the ability to "price upwards when competitors increase their prices" [8]. Informed Repricer advertises a "Smart Price Reset" feature that raises prices during slow sales periods, stating that "the price reset causes your competitors to raise their prices as well, leading to higher overall profit margins" [9]. In this market, collusion is openly offered as a product feature.

Prior to recent research, there had been significant debate over whether algorithmic collusion occurred in real-world markets outside of controlled simulations [10, p. 724]. Assad et al. [10] provides the most rigorous causal evidence that these risks are not just theoretical. Studying the staggered adoption of algorithmic pricing (AP) software by gasoline stations across Germany, they estimate that market-wide adoption raises station-level margins by approximately 3.2 cents per liter (roughly 38%) and mean prices by around 6 cents per liter [10, p. 727]. Critically, the effect is zero for monopolist stations and zero in markets with only partial adoption—it is only when all competitors in a local market adopt AP that prices rise substantially. This pattern is inconsistent with better demand forecasting and instead suggests a mechanism of mutual deterrence against raising prices.

The timing of the Assad et al. findings is also revealing. Margins do not rise immediately upon adoption; the increase builds over a year as algorithms gradually learn the equilibrium strategies that sustain higher prices [10, p. 727]. This learning dynamic is the empirical counterpart to the Q-learning collusion that Calvano et al. document in simulation, as discussed in the next section.

2.2 Q-Learning and Tacit Collusion

The question of whether Q-learning algorithms would spontaneously collude was highly contested until recently. Writing in 2018, Schwalbe [11] argued from a technical perspective that algorithmic collusion was difficult to achieve in practice particularly with more than two sellers. In this scenario, sustaining a collusive equilibrium without direct communication is considerably harder [11, p.3]. Schwalbe conceded only that "one cannot completely rule out the possibility that algorithms might be able to tacitly collude" [11, p. 24], dismissing broader concerns as effectively "legal sci-fi" [11, p. 24]. However, later work reviewed in this section largely refutes this skepticism.

Calvano et al. [3] provides the theoretical and computational foundation for this paper's simulation design. They study what happens when two profit-maximizing Q-

learning algorithms are set to compete in a repeated pricing game, with no ability to communicate and no explicit instruction to collude. The result is robust tacit collusion.

Q-learning is a model-free reinforcement learning algorithm in which agents learn optimal strategies through trial and error. At each period, each agent observes the current market prices and selects a price according to an ε-greedy policy. For each period, it has probability ε_t it experiments with a random price and probability $1 - \varepsilon_t$ it exploits the price that has historically yielded the highest return. The agent does not need to know the demand function or the game structure; it simply learns which prices are profitable over time through a reinforcement learning strategy. This is highly advantageous due to its high performance without requiring any training or complex rulesets [12, p. 162].

Calvano et al. find that in more than 95% of simulated markets, Q-learning agents converge to prices significantly above marginal cost, typically reaching 70–90% of the monopoly price [3, p. 3277]. The agents learn a tit-for-tat strategy: if one agent deviates by undercutting the tacit equilibrium price, the other immediately retaliates with lower prices, making the deviation unprofitable. Punishment is swift and consistent enough that "in more than 95 percent of the cases the punishment makes the deviation unprofitable" [3, p. 3282]. In future periods, the seller algorithm adjusts its Q-values to reflect the lower profits from deviating, which reinforces the collusive equilibrium over time.

The collusion is remarkably robust. It persists under stochastic demand shocks, with more than two seller agents [3, p. 3288], and under asymmetric marginal costs [3, pp. 3289–3290]. These conditions typically destabilize human cartels [3, pp. 3288-3289]. This paper uses the Calvano et al. framework directly as the basis for the simulation, extending it to incorporate heterogeneous seller costs, multiple seller counts, and the platform threshold mechanism.

2.3 Platform Design Interventions

Given that algorithmic collusion emerges from the learning process itself, the natural remedy is not to prohibit algorithms but to change the environment in which they learn. Johnson, Rhodes, and Wildenbeest [13] demonstrate this principle rigorously, showing that platform visibility rules can be designed to make collusion an unstable equilibrium for Q-learning agents.

Their key proposal is Dynamic Price-Directed Prominence (DPDP): a rule that gives higher visibility to the seller who most recently reduced prices, and maintains that visibility advantage as long as the seller does not raise prices or get significantly undercut by a rival. Unlike a static price ranking (Price-Directed Prominence, or PDP), DPDP rewards

competitive behavior dynamically and creates a forward-looking incentive to cut prices today in exchange for market share tomorrow. In their simulation, DPDP reduces prices by approximately 19% relative to the no-intervention baseline (compared to only 7% for static PDP) and raises consumer surplus by 27% [13, pp. 1859-1860].

However, DPDP achieves this by steering all demand to the lowest-priced seller, which compresses seller margins to the point of threatening market viability. Johnson et al. acknowledge that when DPDP drives prices to marginal cost, sellers cannot cover fixed costs and may exit, ultimately harming the consumers the mechanism intends to protect [13, p. 1871]. Brero et al. [14] explores learning-based platform interventions using a similar reinforcement learning setup, demonstrating that platforms can train algorithms specifically designed to disrupt collusive pricing patterns [14, p. 5]. While effective in controlled settings, these approaches share the over-compression limitation of DPDP. The study does not evaluate the long-term impacts that may lead to seller exit or market consolidation, which are critical tests of whether the mechanism is viable in practice.

The threshold mechanism proposed in this paper addresses these limitations directly. Rather than rewarding the cheapest seller with all demand, the threshold simply excludes sellers priced above a ceiling, leaving eligible sellers to compete normally within the feasible price space. Sellers below the threshold continue to earn their normal demand share; only above threshold prices are penalized. The platform's participation constraint is preserved by construction, rather than left as an incidental consequence of how aggressively the mechanism compresses prices.

2.4 The Legal Gap and Regulatory Approaches

The empirical evidence and theoretical mechanisms reviewed above raise a fundamental governance question: if algorithmic collusion produces real consumer harm, why can't regulators simply prohibit it? The answer lies in a structural mismatch between existing antitrust doctrine and the nature of tacit algorithmic coordination.

Legal scholars have identified three models under which algorithmic pricing can produce anticompetitive harm. In the *messenger* model, human agents explicitly agree to fix prices via electronic means—the traditional cartel structure that Sherman Act Section 1 is already built to address. In the *hub-and-spoke* model, a single "hub" firm sets prices that "spoke" firms follow without direct communication between [15, pp. 129–130]. This structure has formed the basis for recent litigation, including against the property management software firm RealPage. In this case, the government alledged that RealPage's software enabled tacit collusion among landlords by allowing them to share

information about their prices and occupancy rates. This facilitated a coordinated strategy of raising tenant rents across the market [16]. The third model, *predictable agents*, involves algorithms designed to react to competitor prices in a systematic and foreseeable way. In cases we have previously described, this can enable tacit collusion without any explicit agreement [11, pp. 5-6]. It is this third model that dominates online marketplaces and is also outside the scope of current antitrust law. As a result of its peculiar nature that involves no explicit communication, yet similar harms to traditional collusion, regulators have struggled to find effective legal tools to address it [15, p. 133].

U.S. antitrust law under Section 1 of the Sherman Act requires a "meeting of the minds"—an explicit agreement or concerted action among competitors [17, p. 455]. The law was designed for human cartels in which executives meet, communicate, and agree to fix prices. Algorithmic tacit collusion leaves no such agreement: algorithms typically do not communicate or negotiate. They simply maximize a reward function in a repeated game, and collusion emerges as the equilibrium outcome. Even if there is parallelism in pricing, this is not sufficient to establish a legal violation [17, p. 456]. As MacKay and Weinstein warn, "it seems that if current policy does not change, AP may significantly increase the risk of false negatives," where anticompetitive harm occurs but no legal violation can be proven in court [12, p. 169].

The difficulties are compounded by the complexity of modern pricing algorithms. Proposed remedies such as pre-release licensing, algorithmic auditing, or outright bans on pricing software each face serious practical or normative objections[15, p. 133]. Auditing neural-network-based repricers for collusive intent is technically infeasible at scale with current methods [11, p. 23]. Banning algorithmic pricing itself would eliminate the substantial efficiency benefits that these tools also generate [18, pp. 1363-1364]. More targeted proposals to limiting learning from competitor prices or repricing too frequently face their own difficulties: the risk of chilling market innovation and the practical challenge of enforcing feature-level restrictions [15, pp. 122, 165]. MacKay and Weinstein argue, however, that this regulatory cost may be worth bearing since pricing algorithms are an "extractive innovation." That is, it is a type of technology that transfers wealth from consumers to businesses without any benefit to consumer welfare [15, p. 123].

Mazumdar [17] argues that Section 5 of the FTC Act offers a more promising avenue. Unlike the Sherman Act, Section 5 does not strictly require proof of agreement; it empowers the FTC to address "unfair methods of competition" that fall short of explicit collusion. Mazumdar proposes treating the deployment of certain pricing algorithms as a unilateral "invitation to collude" [17, p. 454]. This concept is already recognized under Section 5, and the FTC has the ability to act on it without additional legislative authority [17, pp.

463-464]. This would allow the FTC to target the algorithm design itself, without needing to establish a conspiratorial meeting of the minds. It would only need to establish that the algorithm's design and deployment created an "actual anticompetitive effect," regardless of whether there was an explicit agreement [17, p. 462].

The repricer marketing materials described above provide precisely the kind of unilateral announcement that could constitute an "invitation to collude" under Section 5. By publicly advertising strategies designed to coordinate prices upward with rivals, these companies may already be making potentially illegal invitations to collude under Mazumdar's proposed framework. While Section 5 offers a legal pathway, enforcement under this theory has not yet rigorously tested in court, and the court precedent remains partly unsettled [17, pp. 473-474]. A primary reason for this is that the FTC has generally avoided more ambitious statutory interpretations of its authority due to past hostile court rulings [17, pp. 452-453].

On the self-regulation side, the historical record from other industries suggests that voluntary action by platforms can be successful when the legal framework is uncertain but external pressure is high. The Motion Picture Association of America (MPAA) established a voluntary film rating system that enables parents to make informed decisions about children's content, addressing public pressure without government-mandated censorship. The Financial Industry Regulatory Authority (FINRA) similarly demonstrates how industry self-governance can maintain market stability and protect consumers while preserving operational autonomy. In both cases, the critical mechanism was offering the industry a credible alternative to more restrictive government intervention. By demonstrating that the harm could be managed internally, the industry deterred public backlash and preempted harsher legislative responses [19]. Online marketplace platforms face an analogous situation: documented consumer harm, growing regulatory scrutiny, and a legal framework that cannot reliably address the specific mechanism of harm. While the conditions that made MPAA- and FINRA-style self-regulation viable are still emerging, they have the potential to become more significant as public awareness of these issues grows.

This regulatory uncertainty, combined with these historical precedents, creates an independent argument for platform self-regulation: a platform that voluntarily implements pricing rules to prevent collusion may reduce its own exposure to regulatory scrutiny, reputational damage, and potential liability as a facilitator of anticompetitive conduct. The theoretical framework developed in Chapter 3 formalizes this incentive through a model in which regulatory and reputational risk explicitly enter the platform's objective function.

3 Theoretical Framework

Online marketplace platforms such as Amazon occupy an unusual position in the economics literature: they are simultaneously price-setters (through listing rules and ranking algorithms), revenue-maximizers (through commissions), and reputation-sensitive intermediaries (dependent on consumer trust). Despite the practical importance of these platforms, there is no known theoretical framework that jointly captures all three roles in a dynamic setting. This framework is therefore constructed from first principles to capture the platform's core trade-off: allowing tacit collusion increases per-transaction revenue but erodes consumer trust and risks driving sellers out of the market.

By understanding these dynamics we can then develop a theoretical model the platform threshold mechanism operates within. Many of the concepts motivate our simulation design in Chapter 4 and are used to interpret the results. The framework proceeds in four stages. We first characterize the market structure and the platform's revenue problem. We then introduce the platform's dynamic learning problem under cost uncertainty and establish convergence of the proportional adjustment rule. Finally, we analyze the conditions under which voluntary adoption of the threshold is incentive-compatible for the platform.

3.1 Market Structure and Seller Survival

We model a continuous, dynamic marketplace in which the platform sets a price threshold τ_t each period. Sellers are heterogeneous: each draws a marginal cost c_i from a continuous distribution $G(c)$ with density $g(c)$ from the range $[c_{\min}, c_{\max}]$. The platform collects revenue through a commission rate $r \in (0, 1)$ and a fixed per-transaction fee $f \geq 0$.

Definition 3.1. Price Threshold Mechanism[1]

At each period t, the platform sets a price ceiling $\tau_t \in [c_{\min}, p_m]$:

- Seller i is *eligible* iff $p_i \leq \tau_t$; ineligible sellers receive zero demand.
- Aggregate demand $D(\tau_t)$ is allocated exclusively among eligible sellers.

Definition 3.2. Platform Revenue Variables

The platform's per-period decision and fee variables are:

- τ_t: the price ceiling set by the platform at time t
- $r \in (0, 1)$: the percentage-based commission rate per transaction
- $f \geq 0$: the fixed fee charged per transaction

A seller participates in the marketplace only if their after-commission revenue covers their marginal cost. Given threshold τ_t, seller i earns at most $(1 - r)\tau_t - f$ per unit sold. Participation therefore requires:

$$(1 - r)\tau_t - f \;\geq\; c_i \tag{3.1}$$

The fraction of sellers satisfying condition (3.1) defines the seller survival rate:

Definition 3.3. Seller Survival Rate

$$S(\tau_t) \;=\; G((1 - r)\tau_t - f) \tag{3.2}$$

$S(\tau_t)$ is the fraction of the seller population with marginal cost below the effective participation threshold $(1 - r)\tau_t - f$.

Since G is a CDF, $S(\tau_t)$ is non-decreasing in τ_t: raising the threshold allows more sellers to participate profitably. Lowering τ_t tightens the participation constraint and shrinks the active seller pool. At the extreme, if $\tau_t = c_{\min}$, the effective revenue a seller receives after paying platform fees, $(1 - r)\tau_t - f$, falls below any seller's marginal cost, and the active seller pool collapses entirely.

3.2 Reputation Dynamics and Platform Objective

Of course, consumer demand on the platform depends not only on current prices but on accumulated consumer trust. We model this through a reputation state variable $\theta_t \in [0, 1]$, which evolves dynamically based on the platform's pricing history.

[1] An earlier version of this mechanism was introduced in [20, pp. 12–13].

Definition 3.4. Reputation Update Rule

The platform's reputation evolves as:

$$\theta_{t+1} \;=\; \rho\,\theta_t \;+\; (1-\rho)\,\Phi(\tau_t) \tag{3.3}$$

where $\rho \in (0,1)$ is a parameter capturing the persistence of consumer loyalty, and $\Phi(\tau_t)$ is the instantaneous trust signal:

$$\Phi(\tau_t) \;=\; \frac{p_m - \tau_t}{p_m - c_{\min}} \tag{3.4}$$

The trust signal $\Phi(\tau_t)$ decreases linearly as τ_t approaches the monopoly price p_m, reflecting the assumption that consumers react proportionally to price inflation. As $\tau_t \to c_{\min}$, $\Phi \to 1$, signifying full trust; as $\tau_t \to p_m$, $\Phi \to 0$, signifying no trust. The reputation update is therefore a weighted average of past reputations and any current shocks to trust. The parameter ρ governs how quickly trust responds to current pricing behavior: a value close to 1 means consumers have long memories, so a single period of elevated prices barely dents accumulated goodwill. A value close to 0 makes trust easily built and destroyed within a few periods.

The platform's single-period expected revenue integrates the threshold price, demand, and reputation:

$$R_t(\tau_t) \;=\; (r\tau_t + f)\cdot D(\tau_t)\cdot \theta_t \tag{3.5}$$

where $D(\tau_t)$ is the aggregate demand at threshold τ_t. The multiplicative structure of R_t captures that reputation does not merely add to revenue—it scales it. A platform with high accumulated trust θ_t earns proportionally more from the same commission terms than an otherwise identical platform whose pricing history has eroded consumer confidence. The platform maximizes its total discounted expected revenue over an infinite horizon, subject to maintaining a viable seller pool:

Definition 3.5. Platform Optimization Problem

The platform solves:

$$\max_{\{\tau_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \delta^t \left[(r\tau_t + f) \cdot D(\tau_t) \cdot \theta_t\right] \tag{3.6}$$

subject to the seller participation constraint:

$$G((1-r)\tau_t - f) \;\geq\; S_{\min} \quad \forall\, t \tag{3.7}$$

where $S_{\min}$ is the minimum seller survival rate required to fulfill orders.

The key feature of problem (3.6) is the discount factor δ: the platform is maximizing the *present value of its entire future profits*, not just current-period revenue. This intertemporal perspective is central to understanding why a rational platform might voluntarily adopt a price threshold. Without any threshold, sellers compete freely and the market settles at the natural equilibrium that emerges from their strategies. As the simulation demonstrates, this is tacitly collusive and substantially above marginal cost. A platform that sets a binding ceiling τ_t accepts a short-run reduction in per-transaction commissions in exchange for compounding long-run gains: sustained consumer trust (a higher θ_t), a stable and competitive seller pool, and reduced exposure to the reputational and regulatory risks formalized in Section 3.5. Whether these long-run gains outweigh the short-run cost is the central question the rest of this framework addresses.

For platforms, raising τ_t increases per-transaction revenue $(r\tau_t + f)$ but erodes reputation θ_t via equation (3.4), reducing future demand. Lowering τ_t builds consumer trust but tightens the seller participation constraint (3.7) and reduces commission income. We formalize these trade-offs in Sections 3.3 and 3.5.

3.3 Dynamic Threshold Learning Under Uncertainty

In a real-world marketplace, the platform cannot directly observe the seller cost distribution $G(c)$, nor can it assume costs are static. Inflation, supply shocks, and production changes mean the seller cost distribution $G_t(c)$ may shift from period to period. The platform therefore cannot solve problem (3.6) with perfect knowledge; it must learn the optimal threshold adaptively from observable signals.

We assume the platform can perfectly observe its own aggregate demand $D(\tau_t)$, its reputation state θ_t, and the realized seller survival count S_t. These are natural observables for any modern platform. $D(\tau_t)$ is recorded from transaction logs, θ_t can be inferred from user retention metrics, S_t is simply the count of active listings priced at or below τ_t, which the platform can likely monitor in real time. However, the platform cannot observe the underlying cost distribution $G_t(c)$ directly. Sellers do not report their marginal costs, and these costs shift with macroeconomic conditions. This motivates a dynamic approach in which the platform adjusts τ_t based on the observed discrepancy between actual and target seller participation, without requiring knowledge of $G_t(c)$.

Definition 3.6. Oversupply Signal and Target Survival

To maintain a safety buffer above the minimum participation requirement, the platform sets a target survival rate $S_{\text{target}} = S_{\min} + \eta$ for some buffer $\eta > 0$. The per-period oversupply signal is:

$$S_O \;=\; S_t - S_{\text{target}} \tag{3.8}$$

A positive S_O indicates more sellers than the target. This indicates that the threshold can be tightened; a negative S_O indicates seller undersupply, meaning the threshold must be relaxed. For example, if the platform targets 3 active sellers ($S_{\text{target}} = 3$) but observes 5 sellers priced at or below τ_t, then $S_O = 2 > 0$: the market is over-supplied and the threshold can be tightened without immediately risking seller exit.

We consider two learning rules for updating τ_t based on S_O.

Model 1: Fixed-Increment Updates. The simplest approach adjusts the threshold by a fixed step $\varepsilon > 0$ in the direction indicated by the sign of S_O:

$$\tau_{t+1} \;=\; \tau_t \;-\; \varepsilon \cdot \text{sign}(S_O) \tag{3.9}$$

This rule is computationally simple and easy to implement, but it behaves like an on/off thermostat: regardless of whether the room is one degree or twenty degrees below target, it applies exactly the same heat (ε). A fixed step size is poorly suited to large exogenous shocks. If $G_t(c)$ shifts sharply upward due to an inflation event, a small ε may not restore the seller pool quickly enough to prevent supply shortages. Conversely, if ε is set large to handle shocks, the threshold will oscillate around the target but never settle near it. The

fixed-increment rule also loses information; by collapsing S_O to its sign, the platform discards the magnitude of the imbalance.

Model 2: Proportional Adjustment. To allow rapid adaptation while preserving convergence, we propose a proportional control mechanism. It is analogous to a modern modulating furnace that dials its output continuously to match the temperature gap, rather than simply switching completely on or off. The adjustment magnitude is proportional to the size of S_O, governed by a learning rate $\alpha > 0$:

Definition 3.7. Proportional Threshold Update[2]

$$\tau_{t+1} = \tau_t - \alpha S_O \tag{3.10}$$

When $S_O \gg 0$, indicating a large seller surplus, the platform aggressively lowers τ, capturing consumer trust gains. As $S_t \to S_{\text{target}}$ and $S_O \to 0$, adjustments decelerate, softly landing near the optimal threshold. If a cost shock drives $S_O \ll 0$, the platform raises τ to replenish the seller pool.

The proportional rule dominates the fixed-increment approach on two dimensions. First, it responds proportionally to the magnitude of imbalance. Large shocks receive large corrections, small deviations receive small corrections. Second, as shown in the next section, it corresponds to gradient descent on a natural cost function, which guarantees convergence to a unique steady state under conditions that rule (3.9) does not satisfy. Therefore, proportional updating is implemented in the simulation.

3.4 Convergence of the Proportional Rule

We now establish that the proportional update (3.10) converges to a unique stable threshold τ^* at which the seller survival rate exactly meets the platform's target. To do so, we reframe the problem as one of minimizing a cost function that measures how far the current seller pool is from the target. This reframing connects the proportional update rule to the well-understood framework of gradient descent, giving us a principled convergence guarantee.

[2] An earlier version of this mechanism was introduced in [20, pp. 14–15].

Definition 3.8. Cost Function

Define the cost function:

$$V(\tau) \;=\; \tfrac{1}{2}\big(S(\tau) - S_{\text{target}}\big)^2 \tag{3.11}$$

$V(\tau) \geq 0$ with equality if and only if $S(\tau) = S_{\text{target}}$.

The squared form is chosen because it is the simplest smooth, non-negative measure of deviation from our target: $V(\tau) = 0$ when the platform has hit its goal. As $S(\tau)$ diverges from S_{target} in either direction, $V(\tau)$ grows quadratically, penalizing large imbalances more heavily. The cost function therefore captures the platform's desire to maintain a stable seller pool around the target level. The factor of $\frac{1}{2}$ is a standard convenience that cancels with the exponent when differentiating.

The gradient of V with respect to τ follows from the chain rule. Differentiating gives:

$$\frac{d}{d\tau}V(\tau) \;=\; \big(S(\tau) - S_{\text{target}}\big) \cdot \frac{d}{d\tau}S(\tau) \;=\; S_O \cdot S'(\tau) \tag{3.12}$$

To evaluate $S'(\tau)$, note that $S(\tau) = G(u(\tau))$ where $u(\tau) = (1-r)\tau - f$ is a linear function of τ. By the chain rule, $S'(\tau) = G'(u) \cdot u'(\tau) = g(u) \cdot (1-r)$, so:

$$S'(\tau) \;=\; (1-r)\, g((1-r)\tau - f) \;\geq\; 0 \tag{3.13}$$

This is non-negative because both $(1-r) > 0$ and $g(\cdot) \geq 0$ as a probability density.

The second derivative of V follows from differentiating (3.12) with respect to τ using the product rule, recalling that $\frac{d}{d\tau}S_O = \frac{d}{d\tau}(S(\tau) - S_{\text{target}}) = S'(\tau)$:

$$\frac{d^2}{d\tau^2}V(\tau) \;=\; \frac{d}{d\tau}\big[S_O \cdot S'(\tau)\big] \;=\; S'(\tau) \cdot S'(\tau) \;+\; S_O \cdot S''(\tau) \;=\; \big(S'(\tau)\big)^2 + S_O \cdot S''(\tau) \tag{3.14}$$

At any point where $S_O = 0$, equation (3.14) reduces to $(S'(\tau))^2 \geq 0$, confirming that every zero of V is a local minimum. When G is convex—which holds for the uniform distribution used in the simulation— $g'(\cdot) \geq 0$, so $S''(\tau) = (1-r)^2 g'((1-r)\tau - f) \geq 0$ everywhere, making V globally convex and the minimum unique.

Theorem 3.1. Convergence to Steady State

Under the proportional update rule (3.10), a steady state τ^* satisfies $\tau^* = \lim_{t\to\infty} \tau_t$ if and only if $S(\tau^*) = S_{\text{target}}$.

Proof. At steady state, $\tau_{t+1} = \tau_t = \tau^*$. Substituting into rule (3.10):

$$\tau^* = \tau^* - \alpha\, S_O(\tau^*) \tag{3.15}$$

$$0 = -\alpha\, S_O(\tau^*) \tag{3.16}$$

Since $\alpha > 0$, this holds if and only if $S_O(\tau^*) = 0$, which by definition (3.8) is equivalent to $S(\tau^*) = S_{\text{target}}$. □

The proportional update therefore performs a form of gradient descent on $V(\tau)$: each period, the platform takes a step in the direction that reduces the squared imbalance, with larger steps when far from target and smaller steps as it closes in. This is analogous to a hiker descending a valley: the steeper the slope underfoot, the longer the stride; near the bottom, where the ground levels off, steps become shorter. This self-regulating structure is precisely what guarantees convergence: the platform cannot overshoot by a large amount when it is already close to τ^*, because a small S_O produces only a small adjustment. When G is uniform, $S(\tau)$ is linear and $V(\tau)$ is strictly convex, so convergence to τ^* is guaranteed from any initial τ_0.

3.5 Platform Incentive Compatibility

The analysis so far characterizes how the threshold mechanism works and when it converges. A separate question is whether the platform would voluntarily adopt such a mechanism in the first place, given that it reduces short-term commission revenue. We model two sources of external pressure that can make threshold adoption incentive-compatible: reputational risk from consumer defection and regulatory intervention risk.

Definition 3.9. Regulatory Crackdown Probability

Let C_{reg} denote the total cost of a regulatory crackdown (including fines and forced operational changes). The probability of triggering a crackdown in a given period, given maximum price p, is modeled as a convex function of price relative to the competitive benchmark:

$$P_{\text{reg}}(p) \;=\; \left(\frac{p - c_{\min}}{p_m - c_{\min}}\right)^{\beta} \tag{3.17}$$

where $\beta > 1$ governs regulatory sensitivity. The risk is negligible near competitive price levels and accelerates as p approaches the monopoly price p_m. The convex form ($\beta > 1$) reflects the assumption that regulatory attention accumulates nonlinearly: minor markups above marginal cost may go entirely unnoticed, but as prices approach monopoly levels they attract disproportionate scrutiny from enforcers, journalists, and policymakers. A platform pricing modestly above cost faces a very different regulatory environment from one that appears to be extracting monopoly rents.

Incorporating regulatory risk shifts the platform's single-period objective from revenue maximization to risk-adjusted profit maximization:

Definition 3.10. Risk-Adjusted Platform Payoff

The platform's risk-adjusted expected profit in period t is:

$$R_t(p) \;=\; \big[(rp + f) \cdot D(p) \cdot \theta_t\big] \;-\; P_{\text{reg}}(p) \cdot C_{\text{reg}} \tag{3.18}$$

For sufficiently large C_{reg} or β, the marginal cost of additional price inflation ($P_{\text{reg}}(p) \cdot C_{\text{reg}}$) may exceed the marginal revenue gain. This means that the risk-adjusted optimum p^* will be strictly lower than the unconstrained revenue-maximizing price. In this regime, voluntary adoption of the threshold mechanism is incentive-compatible: the platform prefers to self-regulate rather than absorb the expected regulatory cost.

Reputational risk operates through the same channel. If high-price equilibria erode consumer trust, the effective C_{reg} analog for reputation loss can be similarly impactful on profits. Whether regulatory or reputational pressure dominates, the model predicts that platforms facing external scrutiny have an incentive to voluntarily implement price threshold mechanisms. This result is taken up in the policy discussion in Chapter 5, where

the simulation's estimated revenue cost of threshold adoption is used to predict the level of regulatory or reputational exposure required for real-world incentive compatibility.

Note on simulation scope: The reputational risk channel and the regulatory crackdown probability $P_{\text{reg}}(p)$ introduced in this section are part of the full theoretical framework but are *not* directly modeled in the simulation. The Q-learning experiment tests whether the threshold mechanism reduces tacit collusion under controlled conditions, deliberately abstracting away from the external pressures that would motivate voluntary adoption. This separation is intentional: the simulation establishes the *effectiveness* of the threshold mechanism, while the theory here establishes the platform's *incentive* for voluntary adoption. Real-world self-regulation would likely require exactly the kind of external pressure modeled here that make a short-run revenue cost worthwhile for long-run gains.

4 Simulation Design

The simulation follows the Q-learning framework introduced by Calvano et al. [3], extended to incorporate a platform-operated price threshold mechanism and heterogeneous seller costs. Each experiment trains N Q-learning seller agents over 20 million market periods. Two demand specifications are evaluated independently: logit demand and winner-take-all (WTA) demand. For each seller count and demand model, a control run (no threshold) and a treatment run (threshold active) are conducted with identical agent initializations and cost draws, allowing direct comparison of outcomes.

4.1 Q-Learning Setup

Each seller $i \in \{1, \ldots, N\}$ is modeled as a tabular Q-learning agent operating over a discrete action space of $K = 15$ evenly spaced prices on the grid $\mathcal{P} = \{p_1, \ldots, p_K\} \subset [0.05,\ 1.00]$. At each period t, agent i observes the current market state s_t and selects a price p_i^t using an ε-greedy policy: with probability ε_t the agent selects a uniformly random price (exploration), and with probability $1 - \varepsilon_t$ it selects the price with the highest Q-value for s_t (exploitation).

State space: The state s_t encodes the joint price indices of all N sellers along with a discretized bin index for the current threshold τ_t (10 bins). These are combined into a single integer via mixed-radix encoding, yielding a flat Q-table of size $K^N \times N_\tau$ rows, one per state-action pair (s, a). Each agent maintains its own Q-table.

Q-table update: After each period, agent i receives profit π_i^t and updates its Q-table following Calvano et al. [3, p. 3271]:

$$Q_i(s_t,\, a_t^i) \;\leftarrow\; (1-\alpha)\, Q_i(s_t,\, a_t^i) + \alpha \Big[\pi_i^t + \delta \max_{a'} Q_i(s_{t+1},\, a') \Big] \tag{4.1}$$

where $\alpha = 0.10$ is the learning rate and $\delta = 0.95$ is the discount factor. The profit for agent i in period t is:

$$\pi_i^t = (p_i^t - c_i - f) \cdot D_i^t \tag{4.2}$$

where c_i is seller i's marginal cost, $f = 0.01$ is the fixed per-transaction fee paid to the platform, and D_i^t is seller i's demand allocation.

Optimistic initialization: Following Calvano et al., all Q-values are initialized to the expected long-run monopoly payoff $Q_{\text{init}} = \pi_m/(1-\delta)$, where π_m is the per-seller profit at the symmetric monopoly price under logit demand at mean cost $\bar{c} = 0.10$. The monopoly price p_m is computed numerically by maximizing symmetric per-seller logit profit at $\bar{c}$.

Sensitivity to initialization: The choice of Q-table initialization is not merely a convention: it determines which equilibrium the agents converge to. To verify this, we ran an auxiliary experiment comparing three initialization strategies over the full 20 million periods under the 5-seller WTA specification: (i) monopoly-price initialization ($Q = Q_{\text{init}}$, the default), (ii) zero initialization ($Q = 0$), and (iii) uniform-random initialization ($Q \sim \mathcal{U}[0,\ Q_{\text{init}}]$). The results are shown in Figure 4.1; full numerical results are tabulated in Appendix A.6.

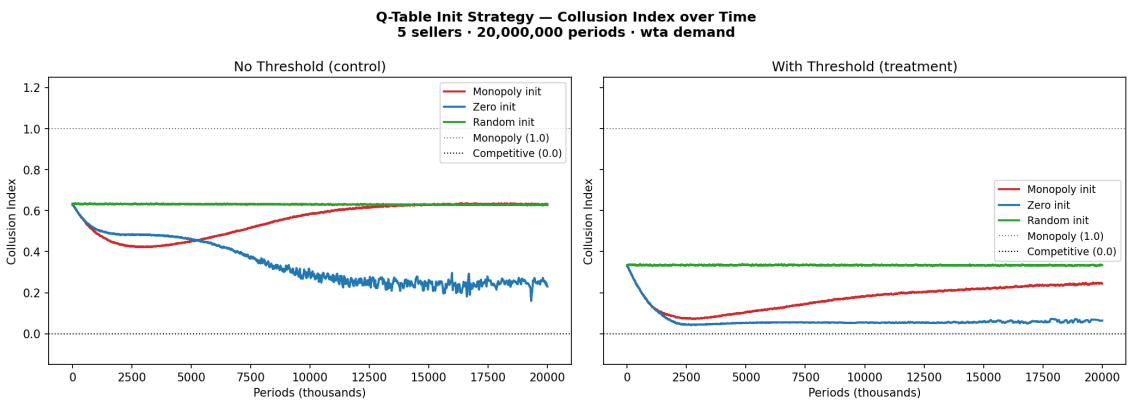

Figure 4.1: Collusion index convergence under three Q-table initialization strategies (5-seller WTA, 20M periods). Monopoly and random initialization track each other closely and converge to the collusive equilibrium (CI $\approx$ 0.63); zero initialization stays at a competitive equilibrium (CI ≈ 0.25) permanently.

Monopoly and random initialization converge to nearly identical control collusion indices (0.632 and 0.629 respectively), confirming that the optimistic choice is not uniquely responsible for collusion. Any initialization with sufficient Q-value variance reaches the same high-price equilibrium. Zero initialization, however, converges to a substantially lower equilibrium (CI $= 0.249$) even at 20 million periods. Starting from $Q = 0$, agents explore competitively from the outset and become anchored to low-price strategies before discovering that mutual restraint is profitable; they never escape this basin. The collusive

equilibrium that the threshold mechanism is designed to disrupt simply does not form under zero initialization.

This result reveals a significant methodological limitation in the Calvano et al. framework. Their central claim is that Q-learning agents autonomously learn to collude [3, p. 3267]. Yet zero initialization converges to a low-price equilibrium even after 20 million periods. Collusion under Q-learning is therefore not a fully emergent property of the algorithm, but of the initial condition: agents initialized near monopoly prices maintain coordination, while agents starting from zero anchor to competitive strategies and never escape that window. The Calvano result is better characterized as showing that if agents *begin* with optimistic expectations of mutual restraint, they sustain it. This is a substantially weaker claim than that Q-learning discovers collusion autonomously as the paper claims. While Calvano et al. claims that the initialization choice does not impact the result of profit gains [3, pp. 3291-3292], our findings in Figure 4.1 suggest otherwise.

This limitation carries through to the results of this paper. Our threshold experiments are conducted against a collusive equilibrium that is itself initialization-dependent, so the policy-relevant question whether real-world pricing algorithms are deployed in a collusively initialized state cannot be settled by simulation alone. Nonetheless, testing the threshold under worst-case conditions is valuable for demonstrating that our mechanism can disrupt collusion even when agents are explicitly biased toward it.

Exploration decay: Exploration probability decays exponentially from $\varepsilon_0 = 1.0$ to $\varepsilon_T = 0.005$ over $T = 20{,}000{,}000$ periods:

$$\varepsilon_t = \varepsilon_0 \cdot \left(\frac{\varepsilon_T}{\varepsilon_0}\right)^{t/T} \tag{4.3}$$

Exponential decay is used in preference to linear decay to avoid the sharp behavioral phase transition that linear schedules introduce near the decay endpoint.

Seller costs: Seller marginal costs are drawn once per full experiment from a uniform distribution: $c_i \sim \mathcal{U}[0.05,\ 0.15]$ with a fixed random seed. The N-seller configuration uses the first N draws, so the 2, 3, and 5-seller runs share costs for their common sellers, making cross-configuration comparisons directly interpretable.

Hyperparameters. Table 4.1 summarizes the key simulation parameters. Many of these are influenced by the decisions of Calvano et al. [3] to ensure comparability, while others were tuned to achieve stable convergence with minimal volatility.

Table 4.1: Simulation hyperparameters.

Parameter	**Symbol**	**Value**
Training periods	T	20,000,000
Price grid levels	K	15
Price grid range		$[0.05,\ 1.00]$
Q-learning rate	α_Q	0.10
Discount factor	γ	0.95
Initial exploration	ε_0	1.0
Final exploration	ε_T	0.005
Seller cost distribution	$G(c)$	$\mathcal{U}[0.05,\ 0.15]$
Mean cost reference	$\bar{c}$	0.10
Fixed platform fee	f	0.01
Commission rate	r	0.05
Threshold learning rate	α	0.05
Threshold drift	δ	0.0005
τ discretization bins	N_τ	10
Logit quality parameter	a	0.5
Logit substitutability	μ	0.25
Outside option utility	a_0	0.0

4.2 Demand Models

Two demand models are implemented and evaluated independently. In both cases the total market demand in a given period is determined by the logit inclusive value (see Section 4.2.1), ensuring that consumer participation is comparable across specifications.

4.2.1 Logit Demand

Under the logit specification, each seller i earns a demand share given by:

$$D_i = \frac{\exp\left(\dfrac{a - p_i}{\mu}\right)}{\exp\left(\dfrac{a_0}{\mu}\right) + \displaystyle\sum_{j=1}^{N} \exp\left(\dfrac{a - p_j}{\mu}\right)} \tag{4.4}$$

where $a = 0.5$ is the product quality parameter, $\mu = 0.25$ governs substitutability (lower μ implies closer substitutes), and $a_0 = 0.0$ is the outside option utility. All sellers earn

positive demand simultaneously, with shares depending continuously on relative prices. This specification follows Calvano et al. [3, p. 3273] directly.

When the threshold is active, a *restricted logit* is applied: sellers priced above τ_t are removed from the choice set and receive zero demand. The remaining eligible sellers split demand via equation (4.4) computed over the eligible subset only. Sellers above the threshold earn zero revenue for that period, so their Q-values for superthreshold prices are driven toward zero across training.

Consumer surplus as follows by measuring the expected utility of the best available option under a discrete choice model:

$$\mathrm{CS}_t = \mu \cdot \log\left(1 + \sum_{j=1}^{N} \exp\left(\frac{a - p_j}{\mu}\right)\right) \tag{4.5}$$

This is the standard log-sum welfare formula derived by Small and Rosen [21, p. 127]. Under a logit model, expected consumer surplus equals μ times the log of the inclusive value, summing the exponentiated net utilities of all inside goods plus the outside option. The measure increases monotonically as prices fall and tends to zero when all prices are far above a.

4.2.2 Winner-Take-All Demand

Under the WTA specification, the eligible seller with the lowest price captures the entire market demand for that period. When multiple sellers tie for the lowest price, one is selected uniformly at random. The total demand captured is equal to $\sum_i D_i$ from equation (4.4) evaluated at the current price vector, ensuring comparability with the logit case.

In the no-threshold case, the lowest-priced seller among all N sellers wins. When the threshold is active, the lowest-priced seller among those priced at or below τ_t wins; if no seller is eligible, no sale occurs.

Consumer surplus under WTA is measured as the buyer surplus from the winning transaction:

$$\mathrm{CS} = \max(0,\ a - p_{\text{winner}}) \tag{4.6}$$

where $a = 0.5$ is the representative consumer's valuation (matching the logit quality parameter) and p_{winner} is the transaction price. This is the standard measure of surplus where the consumer gains the difference between willingness-to-pay and the price actually paid. If no eligible seller exists, the surplus is zero.

4.3 Platform Threshold Mechanism

In the treatment condition, the platform operates the proportional threshold adjustment rule derived in Definition 3.7. At each period, the platform observes the number of eligible sellers S_t (those priced at or below τ_t) and computes the oversupply signal $S_O = S_t - S_{\text{target}}$. The threshold is then updated as:

$$\tau_{t+1} = \text{clip}(\tau_t - \alpha \cdot S_O - \delta,\ c_{\min},\ p_{\max}) \tag{4.7}$$

where $\alpha = 0.05$ is the proportional learning rate, $\delta = 0.0005$ is a constant downward drift applied each period, and the clip ensures τ remains within the valid price range $[c_{\min},\ p_{\max}]$. The drift term prevents the threshold from stalling at a fixed level when $S_O \approx 0$: even at the target seller count, the platform continues "nudging" the ceiling downward. The threshold is initialized at $\tau_0 = p_{\max} = 1.00$, a permissive starting value that imposes no constraint in early training.

4.4 Experimental Configurations and Metrics

Seller counts and target survival. Simulations are run for $N \in \{2, 3, 5\}$ sellers. The target number of eligible sellers S_{target} is set per configuration as shown in Table 4.2. The 2-seller case sets $S_{\text{target}} = N$, meaning the threshold must accommodate all sellers and cannot engage in competitive exclusion; this configuration is included for completeness but is structurally distinct from the 3- and 5-seller cases where $S_{\text{target}} < N$.

Table 4.2: Target eligible sellers by configuration.

Sellers (N)	$S_{\textbf{target}}$	$N - S_{\textbf{target}}$
2	2	0
3	2	1
5	3	2

Metrics: The following metrics are computed as averages over the final 10% of training periods (the last 2,000,000 periods), after agents have substantially converged:

- **Average seller price** ($\bar{p}$): mean price across eligible sellers in the threshold case; mean price across all sellers in the control case. When no sellers are eligible in a given period, $\bar{p}$ falls back to the current τ_t to avoid upward bias from ineligible sellers.

- **Collusion index** (CI): measures how far average prices sit between the competitive and monopoly benchmarks:

$$\text{CI} = \frac{\bar{p} - \bar{c}}{p_m - \bar{c}} \tag{4.8}$$

  A value of 0 corresponds to competitive pricing at mean cost; a value of 1 corresponds to the symmetric monopoly price. Values above 1 are possible when heterogeneous costs push some sellers above the mean-cost monopoly benchmark.

- **Consumer surplus** (CS): logit inclusive value (logit demand) or mean per-period buyer surplus $a - p_{\text{winner}}$ (WTA demand), as defined in Sections 4.2.1 and 4.2.2.

- **Platform revenue**: $(r \cdot p_{\text{winner},t} + f) \cdot D_t$ per period, where $r = 0.05$ is the commission rate and $f = 0.01$ is the fixed fee.

- **Total seller profit**: $\sum_i (p_i^t - c_i - f) \cdot D_i^t$ summed across all sellers per period.

- **Threshold convergence**: mean and standard deviation of τ_t over the final 10% of periods, used to assess stability of the platform's learning rule.

5 Results and Discussion

All results are averaged over the final 10% of training (periods 18,000,001–20,000,000), after agents have substantially converged. Each treatment run is paired with an identical control run (same seller costs, same random seed) with no threshold active. The 3 and 5-seller configurations are the primary focus; the 2-seller case is discussed separately in Section 5.5 and its plots are included in Appendix A.

5.1 Baseline: Collusion Without a Threshold

Under both demand models and all seller counts, Q-learning agents converge to prices substantially above marginal cost, replicating the tacit collusion result of Calvano et al. [3].

Under logit demand, the collusion index reaches 0.710 with 3 sellers and 0.681 with 5 sellers, meaning average prices sit 68–71% of the way between the competitive and the monopoly price. Under WTA demand, collusion indices of 0.620 (3 sellers) and 0.632 (5 sellers) emerge, only modestly below the logit values. This is a notable finding: standard intuition predicts that the winner-take-all punishment structure should destabilize collusion, yet Q-learning agents sustain near-monopoly pricing under both specifications. The WTA collusion index is lower and exhibits higher price volatility (avg price std dev 0.016 vs. 0.002 for 3-seller logit), indicating a less stable yet still collusive equilibrium.

The 5-seller logit control average price (0.557) exceeds the 3-seller (0.522) in absolute terms, yet the 3-seller collusion index is higher (0.710 vs. 0.681), as the larger seller population also raises the monopoly benchmark, compressing the relative collusion measure. The control collusion index is also slightly above 1.0 in the 2-seller logit case (1.031), because heterogeneous costs push some sellers above the mean-cost monopoly benchmark used as the competitive reference.

The WTA result warrants further explanation. Standard intuition holds that WTA markets disfavor collusion because any unilateral undercut captures *all* demand, making punishment swift and severe. Q-learning agents sustain near-monopoly pricing under WTA despite this through the exploration–exploitation tradeoff: agents with ε-greedy policies cannot commit to instantaneous retaliation. When both agents are in exploitation mode,

the punishment delay gives the colluding equilibrium just enough stability to persist. WTA collusion is therefore weaker yet still significant, consistent with Calvano's broader finding that Q-learning generates tacit collusion across a range of market structures.

That replication is, however, initialization-conditional. The Q-init sensitivity experiment shows that zero-initialized agents converge to a competitive equilibrium, never discovering the collusive fixed point. The collusion documented here is therefore better understood as *sustained* rather than *discovered*: Q-learning agents maintain the high-price coordination they begin with but do not arrive at it from scratch. Our results show that the threshold can disrupt collusion once it has formed, which is the relevant scenario for real-world, already-converged markets.

5.2 Threshold Effects Under Logit Demand

Table 5.1 reports outcomes for the 3- and 5-seller logit configurations with and without the threshold. Convergence plots for both configurations are shown in Figures 5.1 and 5.2.

Table 5.1: Logit demand: control vs. treatment outcomes (mean over final 10% of periods).

	3 Sellers		**5 Sellers**	
Metric	Control	Treatment	Control	Treatment
Collusion index	0.710	0.422 (−40.6%)	0.681	0.261 (−61.6%)
Avg seller price	0.522	0.351 (−32.8%)	0.557	0.276 (−50.5%)
Consumer surplus	0.331	0.415 (+25.2%)	0.456	0.561 (+23.2%)
Platform revenue	0.0213	0.0172 (−19.1%)	0.0224	0.0173 (−22.9%)
Total seller profit	0.2913	0.1181 (−59.5%)	0.2582	0.1004 (−61.1%)
τ mean (std dev)	—	0.545 (0.062)	—	0.413 (0.009)

The threshold reduces the collusion index by 0.288 points (−40.6%) in the 3-seller case and by 0.419 points (−61.6%) in the 5-seller case. Consumer surplus rises by 25.2% and 23.2% respectively, as lower eligible prices directly benefit consumers.

The effectiveness of the threshold scales strongly with seller count. With 5 sellers and $S_{\text{target}} = 3$, the platform excludes 2 sellers per period at steady state, creating meaningful competitive pressure on the eligible set. The 5-seller threshold also converges to a tighter steady state: $\tau^* = 0.413$ with a standard deviation of just 0.009, compared to $\tau^* = 0.545$ (std dev 0.062) for 3 sellers. The tighter convergence reflects the larger supply of sellers, which gives the proportional adjustment rule a more stable signal.

Platform revenue falls by 19.1% and 22.9% respectively under the threshold, as lower transaction prices reduce per-period commission income. Total seller profit falls sharply (-59.5% and -61.1%), driven by the combined effect of lower prices and, in the 5-seller case, systematic exclusion of two sellers from eligibility each period.

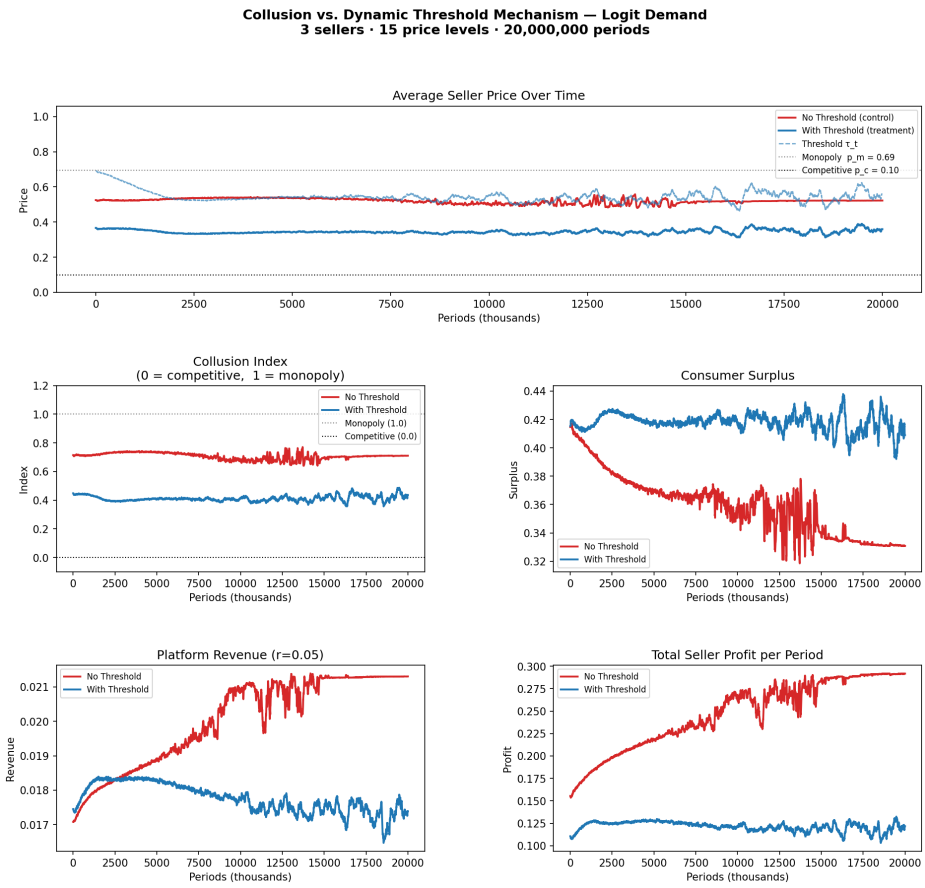

Figure 5.1: 3-seller logit demand over 20 million periods.

5.3 Threshold Effects Under Winner-Take-All Demand

Table 5.2 reports outcomes for the 3- and 5-seller WTA configurations. Convergence plots are shown in Figures 5.3 and 5.4.

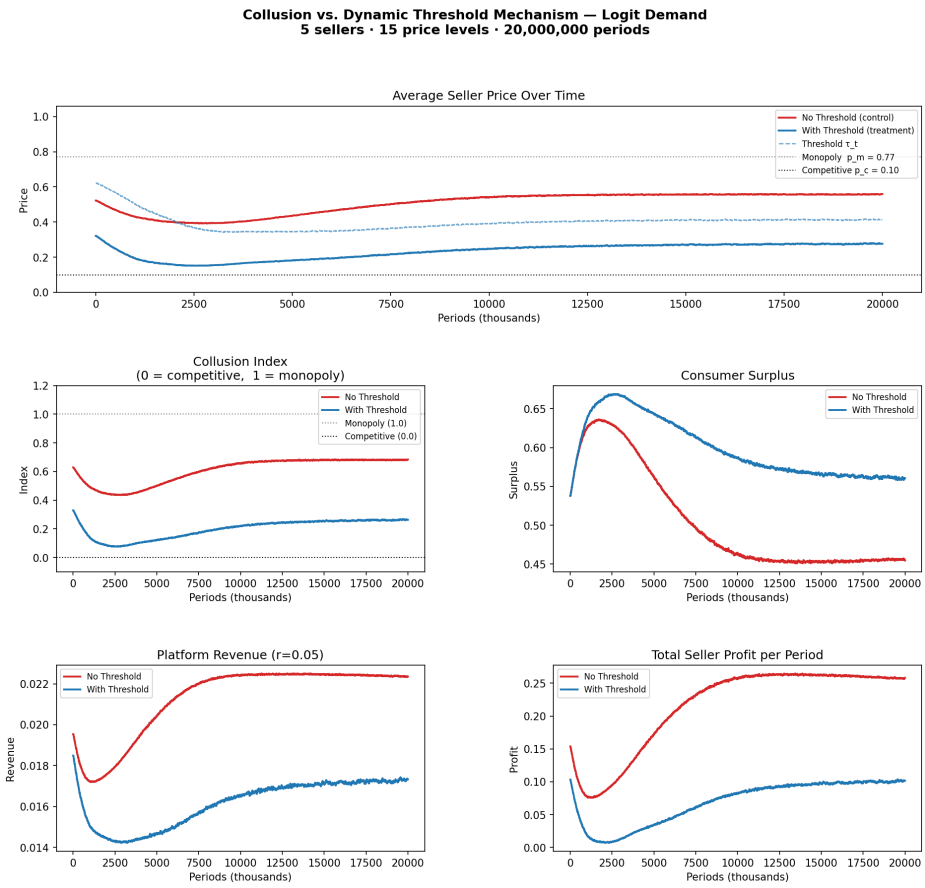

Figure 5.2: 5-seller logit demand over 20 million periods.

Table 5.2: Winner-take-all demand: control vs. treatment outcomes (mean over final 10% of periods). Layout identical to Table 5.1.

	3 Sellers		**5 Sellers**	
Metric	Control	Treatment	Control	Treatment
Collusion index	0.620	0.414 (−33.2%)	0.632	0.242 (−61.7%)
Avg seller price	0.469	0.346 (−26.1%)	0.524	0.262 (−50.0%)
Consumer surplus	0.262	0.243 (−7.2%)	0.279	0.296 (+6.0%)
Platform revenue	0.0175	0.0155 (−11.4%)	0.0179	0.0161 (−9.9%)
Total seller profit	0.0890	0.0749 (−15.8%)	0.0863	0.0604 (−30.0%)
τ mean (std dev)	—	0.625 (0.060)	—	0.514 (0.014)

The collusion index falls by 0.206 points (−33.2%) for 3 sellers and 0.390 points (−61.7%) for 5 sellers. These reductions are comparable in magnitude to the logit case at the 5-seller level, though somewhat smaller at 3 sellers.

The most notable divergence from the logit results is in consumer surplus. Under WTA with 3 sellers, consumer surplus *falls* by 7.2% despite lower average eligible prices. The key is that under WTA the consumer's price is always the *lowest* price in the market, not the average. When all 3 sellers compete, each one is under pressure to undercut the other two, and that three-way race pushes the winning price down. The threshold removes one seller from contention entirely, leaving only 2 competing for the sale. With one fewer rival to undercut, the remaining sellers need not price as aggressively, so the transaction price rises even though individual eligible prices are somewhat lower on average. The loss of competitive pressure dominates, and consumer surplus falls. At 5 sellers the effect reverses: $S_{\text{target}} = 3$ keeps three sellers actively competing, and because those three have been screened to price below τ, their prices are lower than in the unconstrained 3-seller baseline. Thus consumer surplus rises 6.0%.

A further structural feature of WTA results is persistent price cycling. Unlike logit agents, which converge to a stable collusive equilibrium, 3-seller WTA agents exhibit ongoing Edgeworth-cycle-like oscillations throughout the full 20 million training periods, with the rolling standard deviation of the collusion index remaining non-zero and flat rather than decaying. The mechanism mirrors an Edgeworth price cycle: any seller can instantly capture all demand by undercutting by one price step, making a cooperative high-price outcome perpetually tempting to defect from. The Q-learning agents' ε-greedy policy prevents instantaneous retaliation, temporarily sustaining high prices before the next undercutting episode begins anew. This is similar to the pattern Musolff documents in Amazon marketplace data [4]. The WTA collusion indices in Table 5.2 are therefore time-averages over cycles rather than stable equilibrium values. Both control and treatment simulations share this cycling structure, so the threshold comparison remains meaningful: it lowers the *mean* around which prices oscillate, even if it cannot eliminate the cycling itself.

The 5-seller WTA threshold converges tightly ($\tau^* = 0.514$, std dev 0.014), while the 3-seller WTA threshold is less stable ($\tau^* = 0.625$, std dev 0.060), partly reflecting the noisier signal from a single-winner market with fewer sellers. Platform revenue declines modestly (-11.4% and -9.9%), less than under logit, because WTA concentrates revenue on fewer high-demand transactions.

5.4 4-Seller Results: Isolating the Role of S_{target}

The 4-seller experiment holds $N = 4$ fixed and varies $S_{\text{target}} \in \{2, 3\}$, directly testing whether threshold effectiveness is driven by seller count or by the exclusion margin

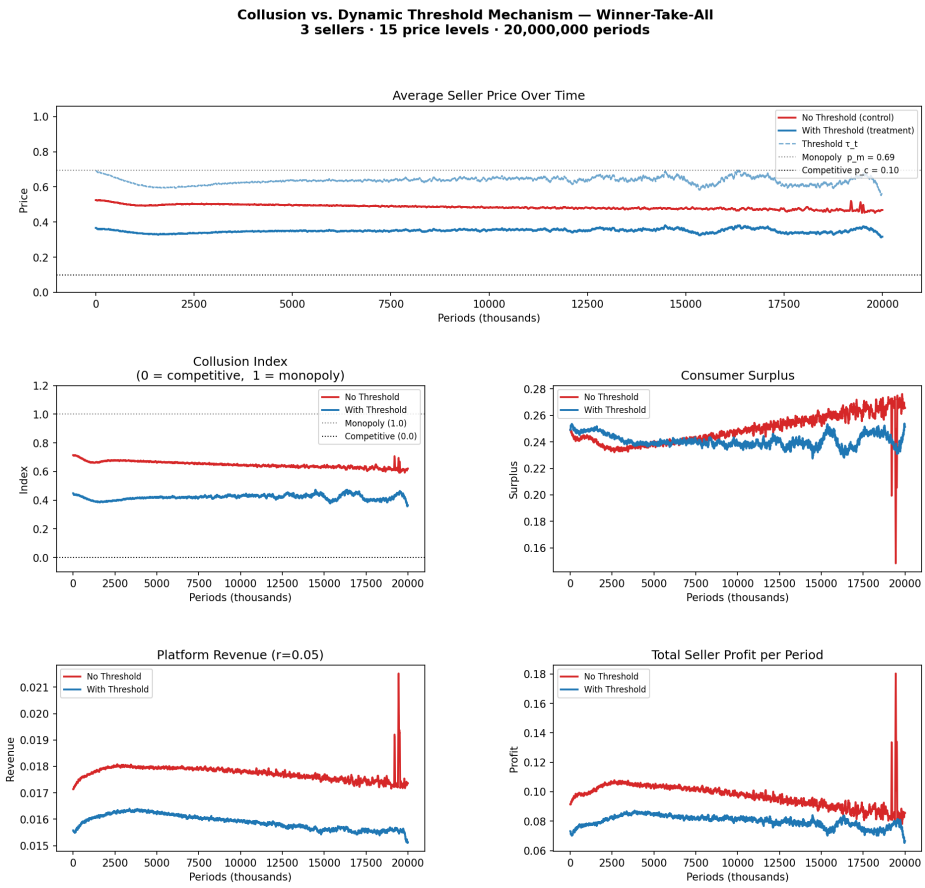

Figure 5.3: 3-seller winner-take-all demand over 20 million periods.

$N - S_{\text{target}}$. Table 5.3 reports outcomes for both target configurations under both demand models.

Table 5.3: 4-seller results under logit and WTA demand, for both S_{target} configurations. Control values are identical across S_{target} since no threshold is active.

		Control	**Treatment**	
Model	**Metric**		$S_T = 2$	$S_T = 3$
Logit	Collusion index	0.692	0.282 (−0.410)	0.449 (−0.242)
	Avg seller price	0.541	0.279 (−48.3%)	0.386 (−28.5%)
	Consumer surplus	0.443	0.482 (+8.7%)	0.471 (+6.2%)
	τ mean (std)	—	0.441 (0.011)	0.660 (0.015)
WTA	Collusion index	0.644	0.283 (−0.361)	0.439 (−0.205)
	Avg seller price	0.510	0.280 (−45.1%)	0.380 (−25.6%)
	Consumer surplus	0.275	0.273 (−0.9%)	0.273 (−0.8%)
	τ mean (std)	—	0.480 (0.011)	0.739 (0.013)

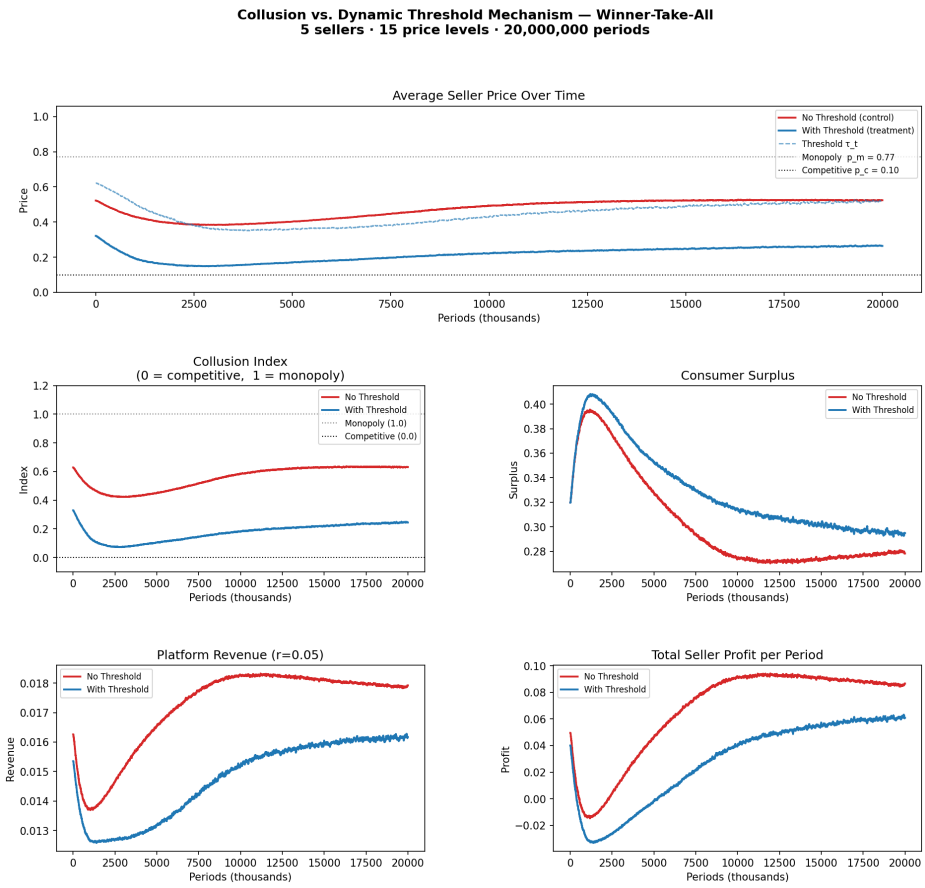

Figure 5.4: 5-seller winner-take-all demand over 20 million periods.

The results are strikingly symmetric. Under logit, $S_{\text{target}} = 2$ ($N - S = 2$) reduces the collusion index by 0.410 points, nearly matching the 5-seller result of -0.419. With $S_{\text{target}} = 3$ ($N - S = 1$), the reduction is 0.242 points, closely matching the 3-seller result of -0.288. The same pattern holds under WTA: $S_{\text{target}} = 2$ yields -0.361 (-0.390 for 5-seller WTA) and $S_{\text{target}} = 3$ yields -0.205 (-0.206 for 3-seller WTA). The τ convergence values mirror this: $S_{\text{target}} = 2$ drives the threshold to approximately 0.44–0.48, matching the aggressive 5-seller convergence, while $S_{\text{target}} = 3$ settles at 0.66–0.74, mirroring the gentler 3-seller convergence. The convergence plots for the $S_{\text{target}} = 2$ configurations are shown in Figures 5.5 and 5.6; the $S_{\text{target}} = 3$ plots are included in Appendix A.3.

5.5 Seller Count and Exclusion Margin Analysis

The 4-seller experiment in Section 5.4 provides a direct test of what drives threshold effectiveness. Table 5.4 reorganizes all configurations by $N - S_{\text{target}}$ — the exclusion margin — rather than by seller count alone.

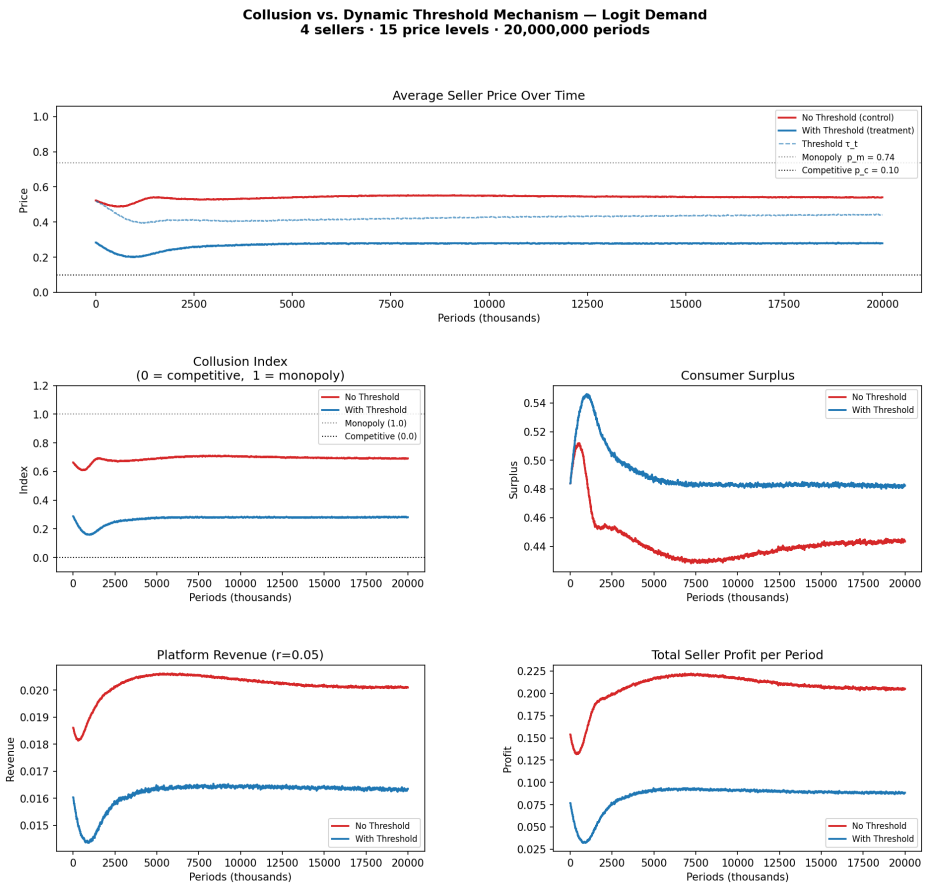

Figure 5.5: 4-seller logit demand, $S_{\text{target}} = 2$ ($N - S = 2$), over 20 million periods. Collusion index falls 0.410 points, comparable to the 5-seller logit result.

Table 5.4: Collusion index reductions grouped by exclusion margin $N - S_{\text{target}}$. Within each group, N and S_{target} vary but $N - S$ is held fixed. Full results in Appendix Table A.1.

$N - S$	N	S_T	**Model**	**CI Control**	**CI Treatment**	**CI**
1	3	2	Logit	0.710	0.422	−0.288
1	3	2	WTA	0.620	0.414	−0.206
1	4	3	Logit	0.692	0.449	−0.242
1	4	3	WTA	0.644	0.439	−0.205
2	4	2	Logit	0.692	0.282	−0.410
2	4	2	WTA	0.644	0.283	−0.361
2	5	3	Logit	0.681	0.261	−0.419
2	5	3	WTA	0.632	0.242	−0.390

Results cluster tightly by exclusion margin, regardless of the specific N or demand model. Among $N - S = 1$ configurations, CI ranges from −0.205 to −0.288 across both demand models and two different (N, S_T) pairs. Among $N - S = 2$ configurations,

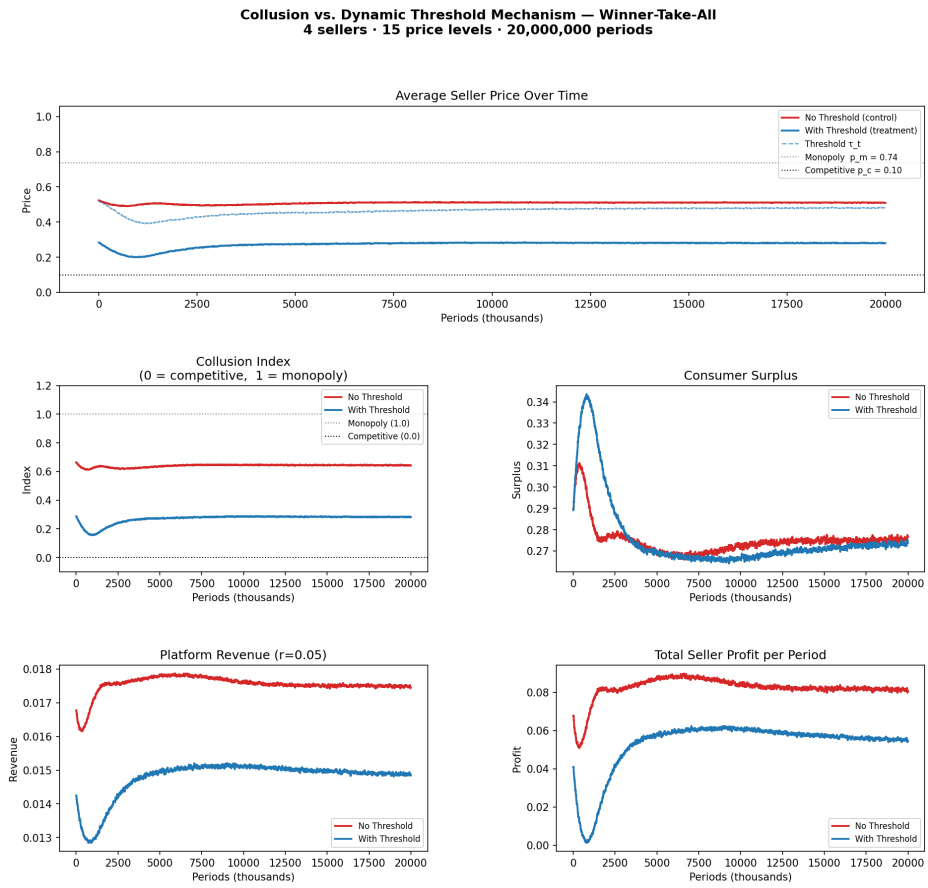

Figure 5.6: 4-seller winner-take-all demand, $S_{\text{target}} = 2$ ($N - S = 2$), over 20 million periods. Collusion index falls 0.361 points, comparable to the 5-seller WTA result.

the range is -0.361 to -0.419. The within-group variation is small; the between-group difference is large and consistent. This supports Theorem I's prediction that the platform's pricing leverage scales with the exclusion margin, not seller count per se. A platform can achieve 5-seller-level collusion reduction with only 4 sellers, simply by setting a more aggressive S_{target}.

When we test the 2-seller case, it is clear that the threshold mechanism fails to function as designed since

$$N \not> S_{\text{target}}$$

The threshold cannot exclude any seller without causing market failure, so it provides no competitive exclusion pressure. Under logit, the collusion index falls only 0.121 points (-11.7%) and the threshold stabilizes at $\tau^* = 0.745$ — well above the colluding price, where it imposes essentially no binding constraint. Under WTA, the threshold *increases* the collusion index by 0.042 points, with τ drifting to 0.926 and never engaging meaningfully

with seller pricing. These outcomes confirm that the mechanism requires $N > S_{\text{target}}$ to function as designed. The 2-seller results therefore represents a structural boundary condition rather than a failure of the threshold mechanism. For the results of the 2-seller case, see Appendix A.2.

5.6 Why the Threshold Reduces Collusion

The threshold mechanism works through two complementary channels: direct compression of the profitable price space and indirect disruption of the coordination equilibrium learned by the Q-learning agents.

In the baseline no-threshold case, Q-learning creates a self-reinforcing coordination equilibrium: each agent's Q-values for high-price states are elevated precisely because competitors also tend to price high, making high prices mutually reinforcing. The threshold breaks this coordination by shifting the entire distribution of profitable actions. By excluding sellers priced above τ from the eligible set, the platform removes high-price actions from the set of profitable strategies. Under logit demand this is especially powerful: ineligible sellers earn zero demand, so any price above τ yields zero reward regardless of what competitors do. Over training, the Q-table values for superthreshold prices are driven to near zero.

The mechanism's effectiveness grows sharply with the exclusion margin $N - S_{\text{target}}$. With more sellers than S_{target}, the platform can credibly exclude the highest-pricing sellers while maintaining a competitive eligible pool. The larger the exclusion margin, the more aggressive this exclusion can be without risking market failure. This explains why the $N - S = 2$ configurations (5-seller and 4-seller $S_T = 2$) produce the largest collusion reductions (approximately 0.36–0.42 points across both demand models) and the tightest τ convergence, while $N - S = 1$ configurations yield more modest reductions of 0.21–0.29 points.

5.7 Demand Model Choice and External Validity

Neither logit nor WTA is a universally correct model of online marketplace demand. Logit is the appropriate specification for markets with *differentiated* products: each seller's demand share depends continuously on their price relative to competitors, reflecting consumer heterogeneity in preferences, brand trust, or fulfillment speed. WTA better

captures *commodity* or *ranked-visibility* markets where a single ranking determines all sales.

Amazon's Buy Box is likely positioned between the two. Since the Buy Box winner is the only visible seller, this would favor WTA as the reference model. However, seller differentiation factors such as reviews and fulfillment reliability mean that consumers do not treat all sellers as identical, introducing logit-like demand splitting. In practice, the Buy Box demand function is closer to a noisy WTA where the lowest-priced eligible seller wins most but not all sales.

This framing has a direct implication for interpreting the simulation results. The threshold mechanism is most effective in markets closer to logit: in commodity markets, welfare gains are smaller and less reliable. The 3-seller WTA case illustrates the risk concretely — consumer surplus fell by 7.2%, reflecting WTA's sensitivity to competitive pressure when the price space is compressed. For a platform like Amazon, where product differentiation is meaningful, the logit results are the more relevant benchmark, and the 23–25% consumer surplus gains under logit are the more applicable estimate of the mechanism's potential.

5.8 Policy Implications

The threshold mechanism and Johnson et al.'s Dynamic Price-Directed Prominence (DPDP) share a common motivation: alter the platform's visibility rules to make high-price equilibria less stable. DPDP achieves this by dynamically rewarding the seller with the lowest price, generating approximately 19% price reductions and 27% consumer surplus gains in their model [13, pp.1850, 1860]. The threshold takes a different approach. Rather than concentrating demand on the cheapest seller, it simply excludes sellers above a ceiling. This distinction matters for market viability. DPDP can over-compress seller profits, risking mass market exit as prices are driven toward marginal cost. Under the threshold, sellers priced below τ remain eligible and earn their normal demand share; only superthreshold prices are penalized. Since the threshold is dynamically adjusted to target a specific number of sellers, it can be calibrated to ensure that the market always has a viable eligible pool.

Complementarity with FTC Section 5. Mazumdar argues that Section 5 of the FTC Act could be used to regulate algorithmic tacit collusion by designating adoption of a collusive pricing algorithms as an "invitations to collude," [17, p. 454]. The threshold mechanism represents a potential solution for platforms to preemptively address such concerns. By

implementing a price threshold, a platform can demonstrate proactive efforts to prevent collusion, potentially reducing regulatory scrutiny and reputational damage.

Whether a platform would voluntarily adopt a threshold depends on whether the regulatory and reputational cost savings outweigh the revenue impact. Definition 3.10 formalizes the regulatory tradeoff: if the probability of regulatory intervention $P_{\text{reg}}(p) = \left(\frac{p-c_{\min}}{p_m-c_{\min}}\right)^{\beta}$ is sufficiently convex in p (high β), the expected regulatory cost $C_{\text{reg}} \cdot P_{\text{reg}}(p)$ grows rapidly with price. Similarly, reputational damage from consumer backlash can be modeled as an increasing function of price, adding further pressure against high-price equilibria. The platform's risk-adjusted profit may therefore be maximized with a threshold that significantly reduces collusion, even if this means sacrificing short-term revenue. The simulation results give this situation empirical grounding: threshold adoption reduces platform revenue by 10–23% across all main configurations, so voluntary adoption is only rational if a platform believes regulatory and reputational costs exceed that range.

6 Conclusion

This paper set out to address a specific gap in the algorithmic collusion literature: while empirical work documents that Q-learning sellers tacitly collude, and theoretical work identifies why existing legal remedies fall short, no framework jointly models the platform's role as both a revenue-maximizer and a potential regulator. We developed such a framework from first principles and used it to evaluate a concrete mechanism through simulation. The implementation of our platform rule shows that a price threshold can significantly reduce algorithmic collusion.

The theoretical framework makes four contributions. The Platform Optimization Problem (Definition 3.5) establishes the platform's core tradeoff between reducing collusion and maintaining a viable seller pool. The Proportional Threshold Update (Definition 3.7) introduces the platform's dynamic learning strategy that does not require knowledge of the seller cost distribution. Theorem 3.1 proves that the threshold rule converges to a unique steady state, and the simulation confirms this. Definition 3.10 formalizes when voluntary adoption of the threshold is incentive-compatible for the platform.

The simulation findings are consistent across demand models and seller counts. Under logit demand, consumer surplus rises by roughly 23–25% alongside large collusion reductions; WTA produces comparable collusion reductions but smaller and less reliable consumer surplus effects, reflecting the winner-take-all structure's sensitivity to the number of sellers competing for a single sale. The key driver of effectiveness is not seller count alone but the exclusion margin $N - S_{\text{target}}$. The 4-seller experiment confirms that a platform with $N = 4$ and $S_{\text{target}} = 2$ achieves virtually identical collusion reduction to the 5-seller case, because in both configurations the platform can credibly exclude two sellers each period.

How far these results transfer to a specific market depends on where that market sits on the logit–WTA spectrum. Amazon's Buy Box is likely between the two: its single-winner visibility favors WTA, but seller differentiation via reviews and fulfillment reliability introduces logit-like demand splitting. For markets with meaningful product differentiation, the logit results are the more applicable benchmark. For near-commodity markets, the WTA results apply.

A recurring concern about threshold adoption is the short-run revenue cost to platforms, which declines by 10–23% in the simulation. The Risk-Adjusted Platform Payoff (Definition 3.10) demonstrates the conditions under which this cost is offset by reduced regulatory and reputational exposure. Unlike Johnson et al.'s Dynamic Price-Directed Prominence [13], which concentrates demand on the cheapest seller and risks driving prices toward marginal cost, the threshold only excludes superthreshold sellers while leaving the eligible pool undisturbed. This design makes the revenue cost more predictable and the mechanism more viable for a platform that must balance competitive pressure against seller participation.

Several limitations constrain the scope of these results. First, the collusive equilibria studied here are initialization-dependent. The sensitivity experiment shows that zero-initialized agents do not discover collusion, even after 20 million periods. The main results therefore test the threshold against a worst-case equilibrium that only arises when agents begin with optimistic expectations; the extent to which real-world algorithms are deployed in such a state is an open empirical question. Second, seller costs are drawn once and held fixed throughout each run; in practice, the cost distribution $G_t(c)$ evolves, and live cost shocks could perturb the convergence of τ. Third, the simulation uses tabular Q-learning agents, which are slow to re-coordinate after disruption. Deep learning agents, with richer state representations and longer memory, might learn to re-coordinate more effectively, potentially weakening the threshold's effect. Fourth, the simulation only models a single platform. In a multi-platform environment, sellers constrained in one marketplace can migrate to others without a threshold, potentially undermining the threshold's effectiveness and changing the welfare calculus. Finally, reputation dynamics (θ_t) are not included in the simulation, so the long-run demand benefit of a competitive reputation remains theoretical. Each of these limitations points to a natural extension for future research.

The field of algorithmic pricing is young and the policy tools available to regulators remain limited. Antitrust doctrine was designed for human cartels and leaves few practical tools for addressing algorithmic behavior. The proportional price threshold developed here represents a different kind of intervention that works within the dynamics of the marketplace rather than against it. It could be effectively implemented by the platform voluntarily or mandated by a regulator to deter supracompetitive pricing. As algorithmic pricing becomes ubiquitous across e-commerce, ride-sharing, and financial markets, these regulatory questions will only grow more pressing. This paper offers one such answer grounded in theory and validated in simulation.

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A Appendices

A.1 Simulation Code

The full simulation code is available on GitHub. The repository includes `config.py` with all hyperparameters, `simulation.py` (the Q-learning simulation), and `plotting.py` (used for all figures and tables).

A.2 2-Seller Results

The 2-seller case sets $S_{\text{target}} = N = 2$, meaning the threshold cannot exclude any seller without eliminating the market. As a result, the platform has no competitive exclusion leverage and the threshold mechanism does not function as designed. These plots are included for completeness but the configuration is not representative of the policy scenario of interest.

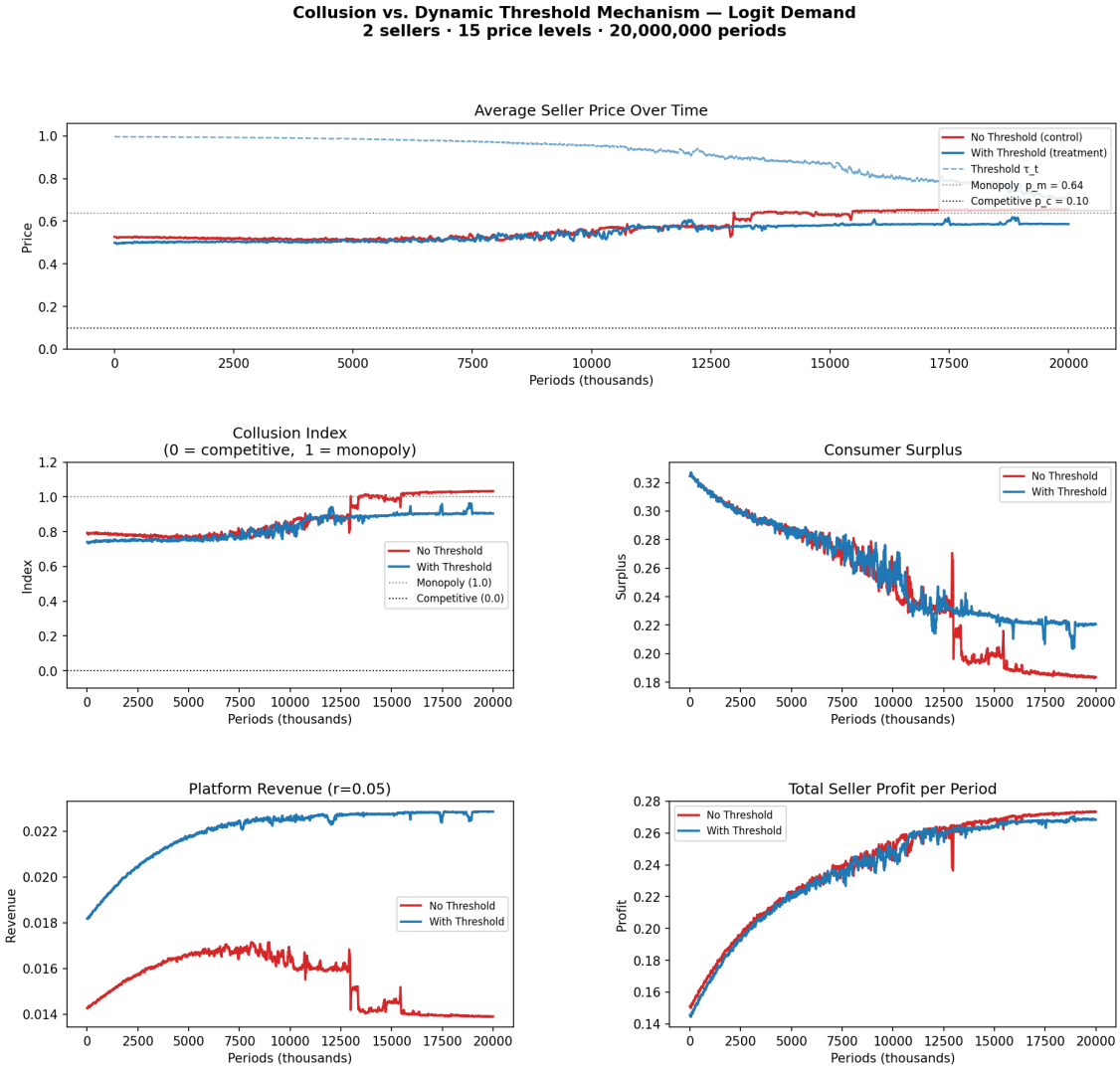

Figure A.1: 2-seller logit demand. The threshold stabilizes at $\tau^* = 0.745$, above the colluding price, and provides only a minimal binding constraint. Collusion index falls 0.121 points (-11.7%).

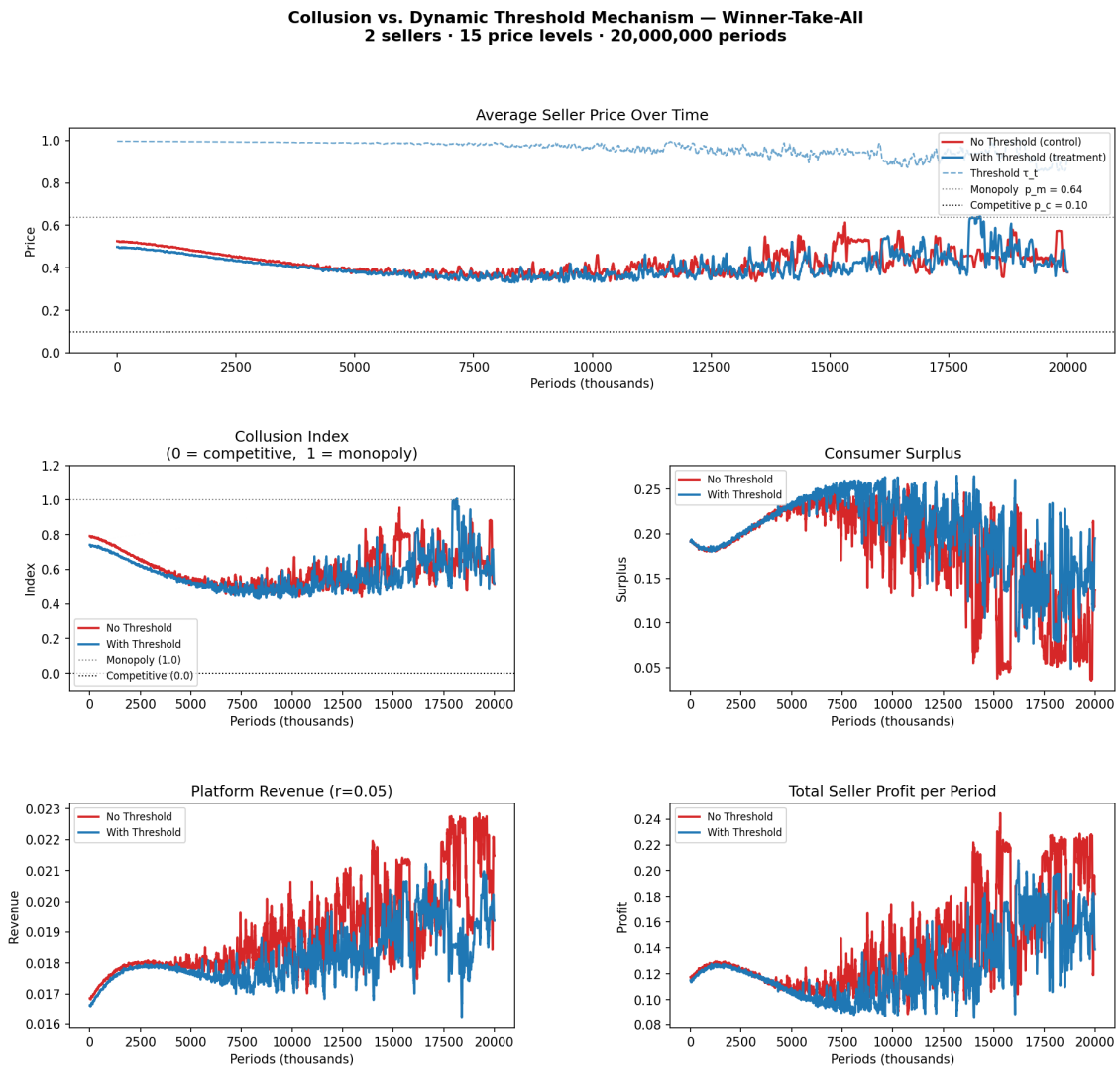

Figure A.2: 2-seller winner-take-all demand. The threshold drifts to $\tau^* = 0.926$ and never engages with seller pricing. The collusion index increases by 0.042 points under the threshold, illustrating the structural boundary condition when $N = S_{\text{target}}$.

A.3 4-Seller Results ($S_{\text{target}} = 3$)

The following plots correspond to the 4-seller, $S_{\text{target}} = 3$ ($N - S = 1$) configuration discussed in Section 5.4. Results are comparable in magnitude to the 3-seller configurations.

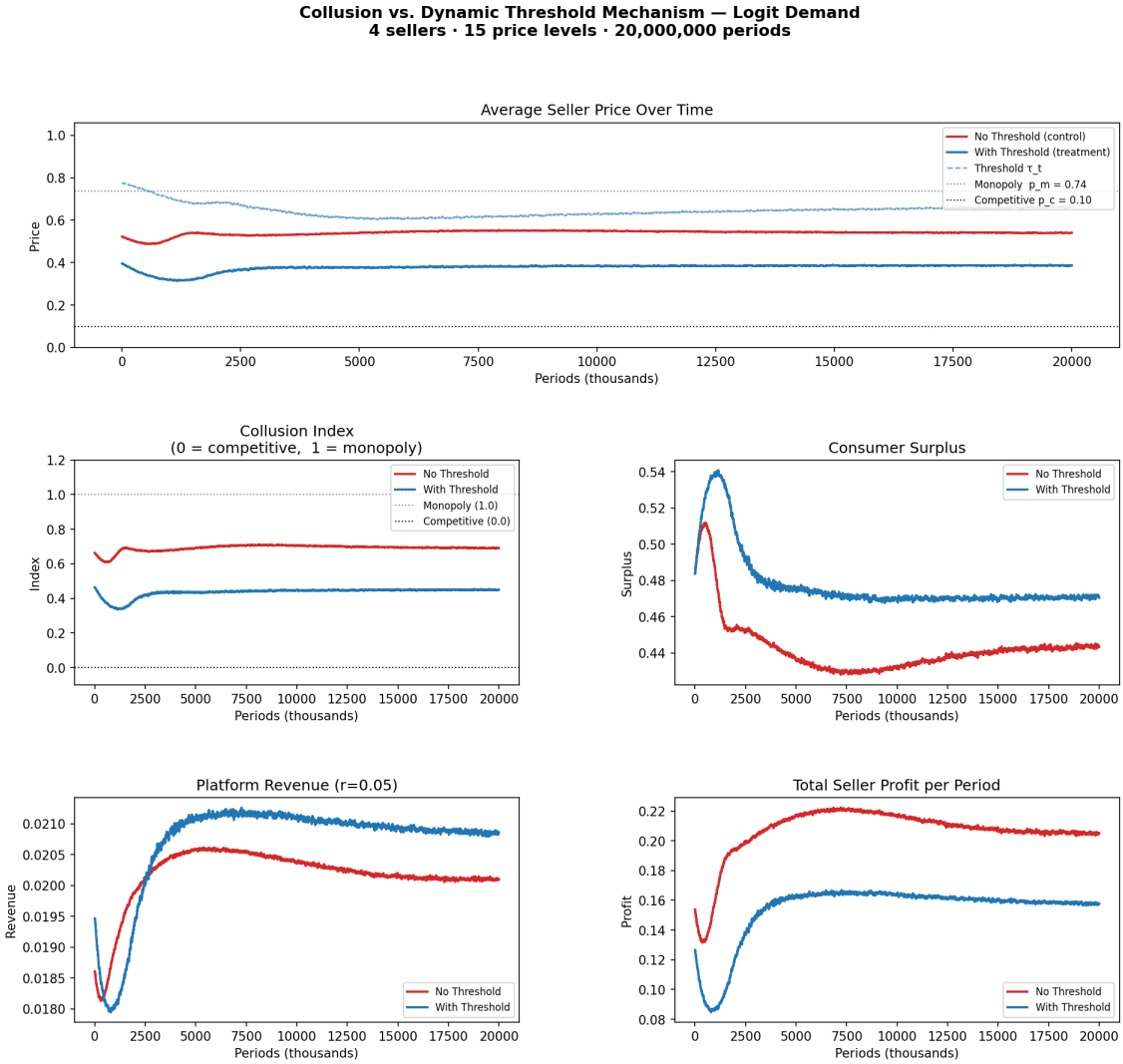

Figure A.3: 4-seller logit demand, $S_{\text{target}} = 3$ ($N - S = 1$). Collusion index falls 0.242 points (-35.0%), closely matching the 3-seller logit result (-0.288).

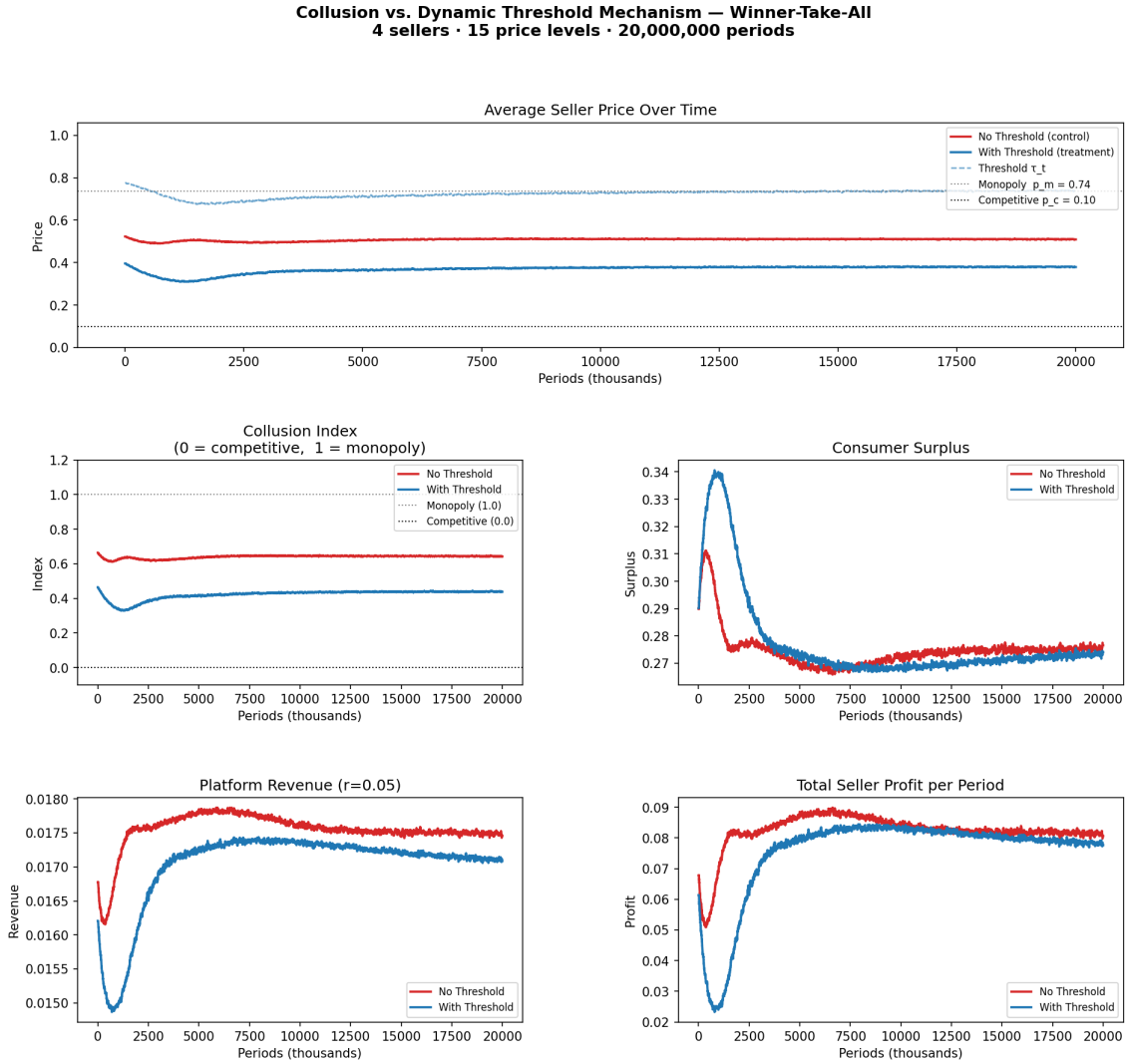

Figure A.4: 4-seller winner-take-all demand, $S_{\text{target}} = 3$ ($N - S = 1$). Collusion index falls 0.205 points (-31.8%), nearly identical to the 3-seller WTA result (-0.206).

A.4 Full Results Table

Table A.1: All configurations: control vs. treatment outcomes (mean over final 10% of periods). CI = collusion index; CS = consumer surplus; PR = platform revenue; SP = total seller profit.

N (S_T)	**Model**	**CI Ctrl**	**CI Treat**	**ΔCI**	**CS %Δ**	**PR %Δ**	**SP %Δ**
2 (S_T=2)	Logit	1.031	0.910	−0.121	+18.9%	+63.8%	−1.6%
2 (S_T=2)	WTA	0.663	0.705	+0.042	+57.7%	−11.5%	−19.9%
3 (S_T=2)	Logit	0.710	0.422	−0.288	+25.2%	−19.1%	−59.5%
3 (S_T=2)	WTA	0.620	0.414	−0.206	−7.2%	−11.4%	−15.8%
4 (S_T=2)	Logit	0.692	0.282	−0.410	+8.7%	−18.8%	−57.0%
4 (S_T=2)	WTA	0.644	0.283	−0.361	−0.9%	−14.9%	−32.0%
4 (S_T=3)	Logit	0.692	0.449	−0.242	+6.2%	+3.8%	−23.0%
4 (S_T=3)	WTA	0.644	0.439	−0.205	−0.8%	−2.1%	−3.8%
5 (S_T=3)	Logit	0.681	0.261	−0.419	+23.2%	−22.9%	−61.1%
5 (S_T=3)	WTA	0.632	0.242	−0.390	+6.0%	−9.9%	−30.0%

A.5 Note on WTA Convergence Plots

Readers may notice a visual discontinuity at the right edge of the 3-seller WTA convergence plots, where the smoothed series appears to spike or flatten abruptly before the end of the x-axis. This is a smoothing boundary artifact: the centered moving-average smoother used in all figures pads the final $w/2$ data points by repeating the last valid smoothed value. When the underlying series is still oscillating at the end of training (as it is for 3-seller WTA) this padding creates a visible flat segment following an apparent spike. The reported summary statistics are computed from raw period averages over the final 10% of training and are unaffected by this artifact.

A.6 Q-Init Sensitivity: Numerical Results

Table A.2: Collusion index by Q-table initialization strategy (5-seller WTA, 20M periods). Control = no threshold; Treatment = threshold active.

Initialization	**Control CI**	**Treatment CI**	**Reduction**
Monopoly (Q_{init})	0.632	0.242	-0.390
Random ($\mathcal{U}[0, Q_{\text{init}}]$)	0.629	0.333	-0.295
Zero	0.249	0.063	-0.186

A.7 Artificial Intelligence Disclosure

With the approval of the faculty advisor, artificial intelligence assistance was used for auxiliary tasks in the preparation of this project. Specifically, Claude Sonnet 4.6 was used to assist with tasks like simulation scripting, LaTeX formatting, and basic summarization. All AI-assisted output was rigorously reviewed for accuracy prior to inclusion.